Assignment 1: WSL, Docker and MySQL Backup & Restore

Objective

Set up a local DevOps environment using WSL and Docker, run MySQL in a container, create a sample database, take a backup, delete and restore the database, and configure persistent storage using Docker volumes.

GitHub Repository

Repository Name: devops-assignment-1-wsl-docker-mysql

Implementation Steps

1. Installed WSL on Windows and configured Ubuntu.
2. Installed Docker inside WSL and verified Docker service.
3. Pulled MySQL Docker image and started container.
4. Created sample database 'company' and inserted employee records.
5. Created database backup using mysqldump.
6. Deleted database to validate recovery.
7. Restored database from backup file.
8. Configured Docker volume for persistent MySQL data storage.

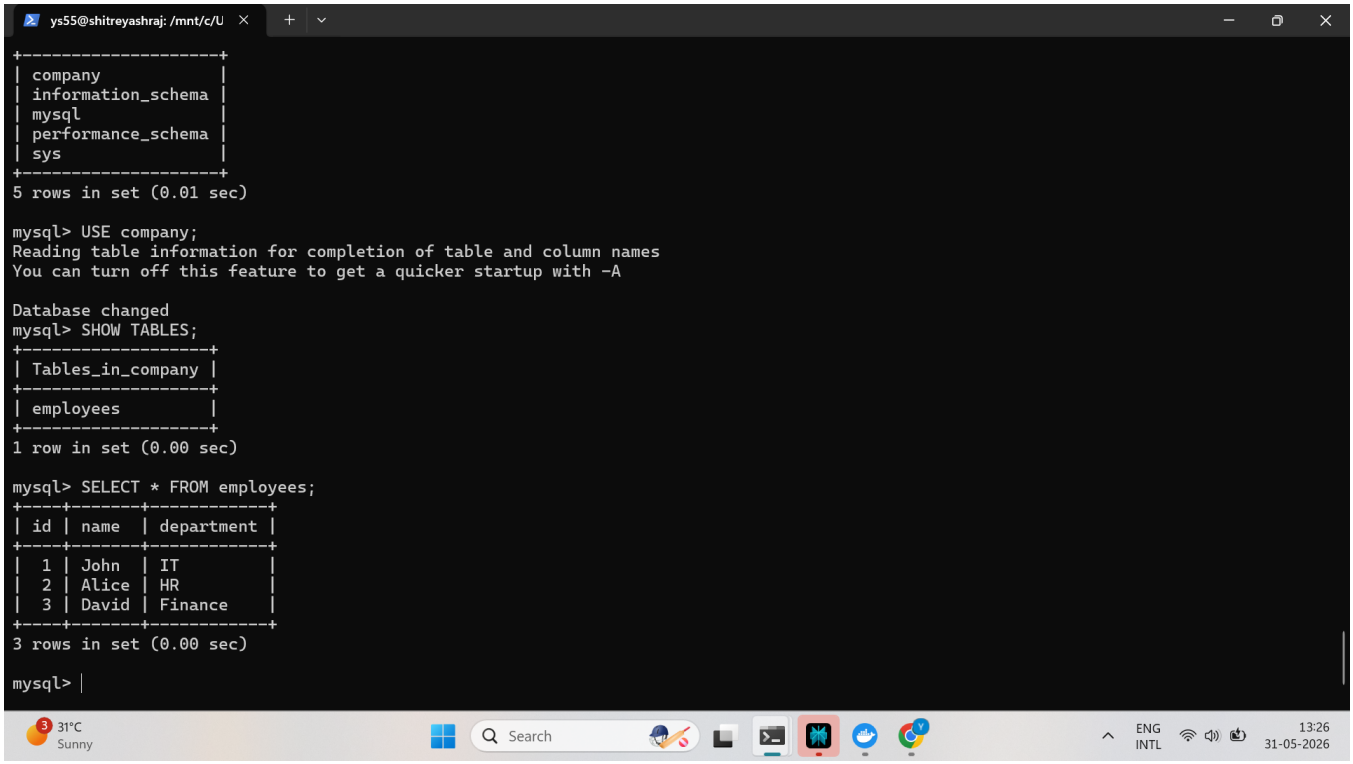
Docker Commands Used

- docker pull mysql:8
- docker run -d --name mysql-db -e MYSQL_ROOT_PASSWORD=root123 -v mysql-data:/var/lib/mysql -p 3306:3306 mysql:8
- docker exec -it mysql-db mysql -uroot -p
- mysqldump -uroot -p company > backup.sql
- mysql -uroot -p company < backup.sql

Volume Configuration

Docker named volume 'mysql-data' was mapped to /var/lib/mysql to ensure persistent database storage.

Screenshot – Successful Database Restoration



Troubleshooting Guide

Issue	Resolution
Docker daemon not running	Restarted Docker Desktop / Docker service
Port 3306 already in use	Stopped conflicting service or changed port
MySQL access denied	Verified root password and restarted container

Rollback / Recovery Procedure

If rollback is required: stop and remove the container, delete volume if needed, and recreate container from saved backup file.

Key Decisions and Configuration Explanation

WSL was used to simulate Linux environment on Windows. Docker containerization simplified setup and cleanup. MySQL volume ensured data persistence even after container restart. Backup and restore was validated using sample employee records.